Sampling Distributions and Standard Errors

Lectures 15–17

QBA201 – Quantitative Business Analysis

Shi Qiu – Fall 2026

A sampling distribution describes a statistic

Repeat the same sampling procedure and calculate the same statistic each time.

The resulting probability distribution describes how that statistic varies across possible samples.

Individual observations and entire sample statistics have different distributions.

One die has a discrete uniform distribution

$$\mathrm{P}(X = x) = 1/6, \qquad x = 1, \ldots, 6.$$

$$\mathrm{E}(X) = 3.5, \quad \mathrm{Var}(X) = 35/12 = 2.9167, \quad SD(X) = 1.7078.$$

Discuss: Can one die show exactly the expected value 3.5?

Two independent dice give 36 possible samples

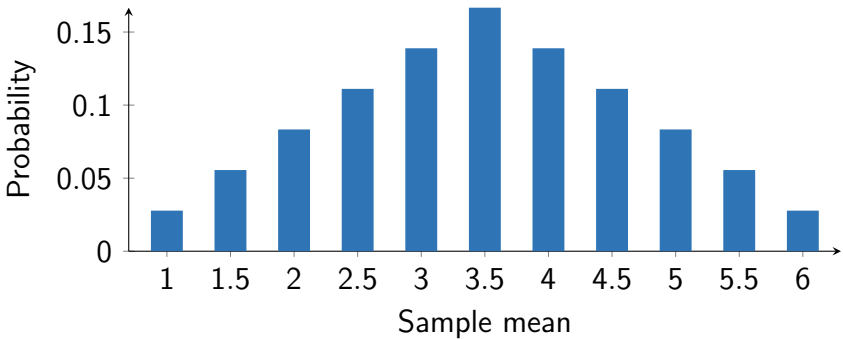
For the ordered pair (X_1, X_2):

$$\bar{X} = \frac{X_1 + X_2}{2}.$$

Sample	Probability	Mean
$(1,1)$	$1/36$	1
$(1,2)$	$1/36$	1.5
$(2,1)$	$1/36$	1.5
$(6,6)$	$1/36$	6

Distinct samples can have the same mean.

The exact distribution of a two-dice mean



The center is 3.5. Variance is $35/24 = 1.4583$.

Averaging preserves the center and reduces spread

Quantity	One die	Mean of two dice
Mean	3.5	3.5
Variance	2.9167	1.4583
Standard deviation	1.7078	1.2076

The variance halves. The SD is divided by $\sqrt{2}$, not by 2.

The general mean and standard-error formulas

For independent, identically distributed observations with finite variance:

$$\mathrm{E}(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The population SD describes individuals. The standard error describes sample means.

The central limit theorem concerns averages

For i.i.d. observations with finite, positive variance:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

approaches a standard normal distribution as n increases.

The original observations need not be normal.

The CLT does not change the population distribution into a normal distribution.

Exact normality and approximate normality

Normal population

An average of independent normal observations is exactly normal for any n.

Nonnormal population

The normal approximation can improve with larger n under the CLT conditions.

Strong skewness or extreme tails can require larger samples.

Standardize using the statistic's spread

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}.$$

1. Identify whether the event concerns one observation or an average.
2. Find the relevant SD or standard error.
3. Standardize the stated cutoff.

Constructed practice: a sample-mean probability

Assume a normal population with $\mu = 80$, $\sigma = 12$, and an independent sample of $n = 36$.

$$\text{SE} = 12/\sqrt{36} = 2.$$

$$\text{P}(\bar{X} > 84) = \text{P}\left(Z > \frac{84 - 80}{2}\right) = 1 - \Phi(2) = 0.0228.$$

A finite population can reduce the standard error

For simple random sampling without replacement from N units:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}.$$

Here σ uses the population-variance divisor N.

Observing a substantial share leaves less sampling uncertainty. The course usually omits this correction when $n/N \leq 0.05$.

The sample proportion is a sample average

Let X count successes among n binary observations.

$$\hat{p} = X/n.$$

Equivalently, code each success as 1 and failure as 0. Then $\hat{p}$ is the average of those indicators.

Discuss: If 72 of 120 sampled customers buy, what is $\hat{p}$?

Match the binomial center and spread

For the count $X \sim \text{Bin}(20, 0.5)$:

$$\mu_X = np = 20(0.5) = 10,$$

$$\sigma_X^2 = np(1-p) = 5, \qquad \sigma_X = \sqrt{5} = 2.2361.$$

Discuss: Why would using 5 as the standard deviation be incorrect?

Check the success and failure counts

The course uses:

$$np \geq 5, \qquad n(1-p) \geq 5.$$

For $n = 20, p = 0.5$, both expected counts equal 10.

For $n = 200, p = 0.01$, $np = 2$, so this rule fails.

A large n alone is not enough when one outcome is rare.

Continuity correction matches one integer bar

For a normal approximation Y to the binomial count X:

$$\mathrm{P}(X = 10) \approx \mathrm{P}(9.5 < Y < 10.5).$$

The interval covers the width of the discrete count's bar.

Use continuity correction when the question calls for it. Follow explicit instructions for sample-proportion approximations.

Check approximation against the exact probability

$$\mathrm{P}(X = 10) = \binom{20}{10}(0.5)^{20} = 0.1762.$$

For Y normal with mean 10 and SD $\sqrt{5}$:

$$\mathrm{P}(9.5 < Y < 10.5) = 0.1769.$$

Mean and standard error of a sample proportion

Under independent Bernoulli sampling:

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}.$$

The probability calculation uses the given population proportion p, not an estimated substitute unless the question asks for one.

Constructed practice: a proportion above a cutoff

Assume $p = 0.60$, $n = 400$, and independent binary observations.

$$np = 240, \quad n(1-p) = 160, \quad \text{SE} = \sqrt{.6(.4)/400} = 0.024495.$$

Without continuity correction:

$$\text{P}(\hat{p} > .65) \approx 1 - \Phi(2.0412) = 0.0206.$$

Two sample means need two sampling models

Take independent random samples from two populations.

If both populations are normal, $\bar{X}_1 - \bar{X}_2$ is normal.

Otherwise a normal approximation can be justified when both sample means are well approximated by normal distributions.

Independent groups and paired observations need different calculations.

Center and spread of the difference

$$\mathrm{E}(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2.$$

For independent samples:

$$\mathrm{SE}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

The variances add even though the statistic is a difference.

A class comparison is not automatically a teaching effect

A difference in section means can reflect:

- Sampling variation.
- Differences in preparation or student selection.
- Differences in measurement or instruction.

Discuss: What would make a comparison of two sections more credible?

The salary-model inputs

	University 1	University 2
Mean salary	62,000	60,000
Population SD	14,500	18,300
Sample size	50	60

Salaries are annual euros. Assume normal populations and independent samples.

Discuss: Translate the event that the first sample mean exceeds the second.

Calculate the probability of a positive observed difference

$$D = \bar{X}_1 - \bar{X}_2, \quad \mathrm{E}(D) = 2,000, \quad \mathrm{SE}(D) = 3,128.3382.$$

$$\mathrm{P}(D > 0) = \mathrm{P}\left(Z > \frac{0 - 2,000}{3,128.3382}\right) = 0.7387.$$

Inference starts with an unknown population quantity

In practice, μ or p is usually unknown.

We observe a sample statistic, then use its sampling behavior to assess plausible population values.

A sampling distribution connects the observed statistic to the unknown parameter.